Equations for NE533

$$Q = E_f \times N_f \times \sigma_f \times \phi$$

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

$$LHR = \pi R^2 Q_{av}$$

$$T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s$$

$$T_0 - T_s = \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k}$$

$$T_s - T_{IC} = \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}}$$

$$T_{IC} - T_{OC} = \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}}$$

$$T_{OC} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}}$$

$$h_{gap/clad} = \frac{k_{gap/clad}}{t_{gap/clad}}$$

$$k_{gap}(He) = 16 \times 10^{-6} \times T^{0.79}$$

$$k_{gap}(Xe) = 0.7 \times 10^{-6} \times T^{0.79}$$

$$k_{gap}(y) = (1 - y) \times k_{He} + y \times k_{Xe}$$

$$LHR\left(\frac{z}{Z_0}\right) = LHR^0 cos\left[\frac{\pi}{2\gamma}\left(\frac{z}{Z_0} - 1\right)\right]$$

$$T_{cool}(z) - T_{cool}^{in} = \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \left\{ sin\frac{\pi}{2\gamma} + sin\left[\frac{\pi}{2\gamma}\left(\frac{z}{Z_0} - 1\right)\right]\right\}$$

$$k_{ox} = \frac{1}{A + BT} \ (W/cmK)$$

$$\frac{1}{B} ln\left(\frac{A + BT_0}{A + BT_s}\right) = \frac{LHR}{4\pi}$$

$$A = 3.8 + 200 \times FIMA \ (cmK/W)$$

$$B = 0.0217 \ (cm/W)$$